

Building a fair member quality score for a small NSFW Discord community

A well-designed scoring system for a sub-100-member server should function as an inactivity detector, not a performance ranking. The strongest research finding across community science, gamification psychology, and platform design is that scoring systems succeed when they identify genuine disengagement and fail when they create competitive pressure. For an NSFW photo-sharing community specifically, the stakes are higher: members who have shared intimate content experience removal as rejection, and surveillance of participation patterns can suppress the vulnerability that makes the community valuable in the first place. This report synthesizes academic research, platform documentation, and practitioner knowledge into a concrete framework for building an ethical, effective scoring algorithm using only Discord bot data.

Your server sits in a scientifically ideal size range

A community of under 100 members occupies what researcher Christopher Allen identified as the optimal zone for creative online groups: **25–80 active members, with 45–50 as the sweet spot.** [Life With Alacrity](#) Robin Dunbar’s nested social layers place this squarely in the “affinity group” tier (~50 people), where members can know each other personally, remember previous conversations, and notice who contributes. Emily Webber and Dunbar’s 2020 research found that communities of ~40 or fewer can operate democratically without formal leadership hierarchies, while larger groups need recognized authority structures.

The decision to cap membership is well-supported. Richard Millington of FeverBee warns that mature communities should never try to artificially increase active membership, [FeverBee](#) and capping creates scarcity that increases perceived value. When members know the server is full and new applicants are waiting, the community signals that belonging here means something. [Influencers Time](#) However, this same scarcity is what makes the scoring system ethically fraught — every removal decision carries weight in a group where **everyone notices who’s gone.**

Critical mass matters too. FeverBee identifies **~50 actively participating members** as the threshold for self-sustaining activity. [FeverBee](#) If your server has 95 members but only 30 are intermittently active, you don’t have a capacity problem — you have an engagement problem that pruning alone won’t solve. Before implementing scoring, establish your

baseline: how many members are genuinely active weekly?

A key insight from micro-community research: **“Most communities do not fail because they are too small. They fail because they are too vague.”** Influencers Time Tight identity and shared purpose matter more than scale. For a photo-sharing server, this means the scoring system should reinforce the community’s identity (mutual sharing, reciprocal engagement, interpersonal connection) rather than just measuring raw activity.

The 1% rule is wrong for small curated communities

Jakob Nielsen’s famous 90-9-1 participation inequality — 90% lurk, 9% contribute occasionally, 1% create — was derived from large, open platforms Wikipedia Grokipedia like Wikipedia Wikipedia (where the top 1,000 editors produce two-thirds of all edits). Nielsen Norman Group It does not describe your server.

Higher Logic’s Engagement Trends Report found that in small communities under 5,000 members, **up to 23% of members actively create content**, Higher Logic with another significant portion responding and reacting. The Community Roundtable’s survey data shows the best communities achieve participation profiles of **17% lurkers, 57% contributors, 26% creators** — nearly inverting the 90-9-1 ratio. Ning Medium Noele Flowers’ 2024 research confirmed this directly: community managers running both large and small communities see the 90-9-1 rule hold for large communities but break down for smaller ones, where **50–60% of members attend monthly programming**. Noele Flowers

For a sub-100 curated server, realistic healthy targets based on aggregated research:

Role	Percentage	Members (of 95)	Behavior
Active creators	15–25%	14–24	Post photos regularly
Active engagers	30–40%	29–38	React, comment, converse
Passive consumers	25–40%	24–38	Read, view, occasionally react
Truly inactive	<10%	<10	Haven’t logged in for months

The scoring system should only target that bottom tier — the genuinely absent. If you’re pruning passive consumers who still log in and view content, you’re cutting the audience that makes posting feel worthwhile. Research consistently shows that lurkers validate contributors: Bettermode when a photo gets seven reactions from quiet members, the poster feels seen. Community Remove those quiet members, and creators lose their reason

What engagement behaviors actually indicate community health

Academic and practitioner research converges on several metrics that predict whether a community thrives or dies. The most important are not raw message counts.

Reciprocity is the strongest health signal. Surma's 2016 study in *Computer Communications* (N=392) found strong empirical evidence that reciprocal engagement — liking others' content, which in turn generates likes on your own — is the core engine of healthy content communities. [Iranarze](#) Social Exchange Theory (Blau, 1964) predicts this: members sustain participation when they perceive a fair exchange of value. In a photo-sharing server, this means the ratio of engagement given to engagement received matters more than the absolute quantity of either. A member who reacts to 30 photos but never posts is a valuable community citizen. A member who posts 30 photos but never reacts to anyone else's is extracting value without reciprocating.

Member-to-member interaction ratio is critical. When the admin or a single power user generates most of the conversation, the community is fragile. Healthy communities show spontaneous peer interaction — members responding to each other without prompting. Track what percentage of replies are member-to-member versus member-to-admin.

Engagement breadth outperforms engagement depth. A member who interacts with 15 different people across multiple channels is more valuable to community health than one who sends 200 messages in a single thread with one friend. Social network analysis research shows that members with high **betweenness centrality** — those who bridge otherwise-disconnected subgroups — are disproportionately important. If they leave, the community can fragment into cliques.

Content diversity and new creator emergence are leading indicators. [Bettermode](#) Track whether lurkers are gradually starting to post (healthy) or whether the same three members produce all the content (fragile). The Flickr group study by Kennedy and Shamma (AAAI ICWSM 2016) found that social features, engagement consistency, and content characteristics predicted group survival with **82% accuracy**. [AAAI](#)

Lessons from existing reputation systems worth stealing

Discourse trust levels offer the best structural model

Discourse's five-tier system is the most thoughtfully designed reputation framework in production. Its key innovations for your context:

- **Reading is the foundational behavior.** Trust Level 1 requires zero posting — only entering 5 topics, reading 30 posts, and spending 10 minutes reading. This deliberately values consumption. [Discourse](#)
- **Trust Level 3 uses a rolling 100-day window** and can be lost. Requirements include visiting 50% of the last 100 days, viewing 25% of recent topics, giving likes on at least 7 different days from at least 5 unique users. This prevents "earn and coast" behavior.
- **Anti-gaming is structural:** requiring likes from unique users on unique days prevents like-trading rings. PMs are excluded from all counts. A 14-day grace period prevents immediate demotion after first achieving a level.
- **Relative thresholds** ("25% of topics created in the last 100 days") scale automatically with community size.

EPGP from gaming solves the veteran-vs-newcomer problem

The Effort Points / Gear Points system from MMO raiding guilds is remarkably applicable. Players earn EP for attendance and spend GP when receiving loot. Priority = EP/GP. The critical feature: **both EP and GP decay by 10–30% weekly**, meaning historical contribution matters but recent participation matters more. A veteran who stops showing up sees their priority drop within weeks, while an active newcomer catches up in roughly 15 raids. The base GP floor prevents new members from having infinite or zero priority.

Private tracker ratio systems model reciprocity enforcement

What.CD and similar invite-only communities required users to maintain upload/download ratios above a minimum threshold (typically ≥ 0.5 –1.0). This enforced reciprocity — you couldn't just consume without giving back. The accountability chain (your inviter is responsible for your behavior) created natural quality filtering. These communities were widely regarded as having the highest content quality standards anywhere online.

Known failure modes to avoid

Goodhart's Law ("when a measure becomes a target, it ceases to be a good measure") is the central risk. Reddit karma farming, Stack Overflow's "fastest gun in the west" phenomenon (racing to answer first rather than best), and badge-chasing behavior (contributions spike before earning a badge, then immediately drop) all demonstrate this.

[ResearchGate](#) Stack Overflow found that **13.8% of high-reputation users earned most of their points from asking questions, not answering them** — the metric rewarded something other than expertise. [Petertsehsun](#)

The Matthew Effect (rich-get-richer) compounds over time unless addressed with decay mechanisms. Without decay, early members accumulate insurmountable leads, and newcomers can never compete regardless of their current contribution quality.

Chilling effects are documented across platforms. Reddit users avoid posting unpopular but valid opinions due to downvote fear. A 2025 PLOS ONE study found social credit scores **reduce interpersonal cooperation and trust**, creating persistent biases that resist correction even when contradicted by evidence.

A concrete scoring algorithm built for your data

The following framework uses only the data your bot already tracks: messages, timestamps, replies, reactions given/received, and attachment types.

Engagement type weights

Weight engagement *given* at least as heavily as content creation. The research consensus is that actions helping others (replying, reacting) are at least as valuable as creating content.

Action	Base points	Rationale
Photo/attachment posted	4	Core community contribution
Reply sent to another member	5	Highest-effort social engagement
Reaction given to others' content	3	Active validation of community members
Text message in conversation	2	Community participation
Reaction received	1.5	Quality signal (earned, not given)
Being replied to	1	Passive indicator of conversation value

Time decay: exponential within a bounded window

Use exponential decay with a **90-day rolling window**. Events older than 90 days are excluded entirely; events within the window decay smoothly so recent activity counts more without a harsh cliff.

$$\text{DecayedScore} = \text{BasePoints} \times 0.5^{(\text{days_elapsed} / \text{half_life})}$$

Recommended half-lives by event type:

Event type	Half-life	Rationale
Reactions given	10 days	Low-effort, frequent; recent ones matter most
Messages sent	14 days	Medium persistence
Replies sent	21 days	Higher effort, longer relevance
Photos posted	21 days	Core content, sustained value
Reactions received	14 days	Quality validation, moderate persistence

A 14-day half-life means an action from two weeks ago is worth 50% of its original points; from a month ago, 25%; from two months ago, ~6%. Medium This naturally prioritizes current engagement without a sharp cutoff.

Consistency bonus using coefficient of variation

Rather than streaks (which create anxiety about breaking them), measure consistency as the inverse of variability:

$$\text{ConsistencyScore} = \text{ActiveWeeks} / \text{TotalWeeks} \text{ (within 90-day window)}$$

A member active 10 of the last 13 weeks scores 0.77. Apply this as a **multiplier of 0.5 to 1.2** on the base score — consistent members get a modest boost, but inconsistency alone doesn't tank a score.

Daily caps to prevent gaming

Following Stack Overflow's 200-point daily limit Stackoverflow and MEE6's XP cooldown pattern, cap scoring events per day:

- Messages: 30 scored per day (roughly one every 30 minutes across 15 hours)
- Reactions given: 25 per day
- Replies: 20 per day
- Attachments: 8 per day

Any activity beyond the cap still "counts" for presence but doesn't add to the score. This prevents spam-grinding while rewarding genuine participation.

Anti-gaming measures

Build these in from the start, following Discourse's structural approach:

- **Require engagement breadth:** Reactions to the same person beyond 3/day score at 50% value; beyond 5/day, zero. This prevents reciprocity pods.
- **Minimum message length:** Messages under 5 characters don't score (prevents "lol" spam).
- **Self-interaction exclusion:** Reactions to your own messages don't count.
- **Exclude bot-command channels** from all scoring.
- **Logarithmic diminishing returns:** The 1st–10th reactions each day are worth full value; 11th–20th worth 50%; 21st–25th worth 25%. This follows Reddit's logarithmic vote scaling principle.

Reciprocity and network health metrics

Beyond the individual score, track these community-level metrics monthly:

- **Engagement Gini coefficient:** How evenly distributed is engagement across members? A coefficient approaching 1.0 means a few members do everything (fragile); closer to 0 means engagement is spread widely (healthy).
- **Reciprocity ratio per member:** Engagement given ÷ engagement received. Members with ratios far below 1.0 (consuming without giving back) are potential removal candidates. Members well above 1.0 are community pillars.
- **Bridging score:** Count how many unique members each person interacts with. Members who engage with 15+ others are high-value connectors; members who only interact with 1–2 people may be in a private friendship rather than participating in the community.

NSFW communities play by different rules

Research on NSFW Reddit communities (Corradini et al., 2021, *Information Sciences*) found that NSFW communities tend to have **stronger in-group cohesion and mutual support than SFW equivalents**. NSFW authors publish more per user, are concentrated in niche communities, and show openness to new members from the start. This is good news for your server — the dynamics naturally favor the tight-knit engagement you want.

But the risks are also unique. A 2025 arXiv study of the top 15 NSFW subreddits found persistent challenges with non-consensual content sharing, throwaway accounts used to evade accountability, and "social springboard" behavior where members use the community purely to redirect audiences to monetization platforms. [arXiv](#) [ResearchGate](#) ACM research on OnlyFans creators (2025) identified boundary erosion, doxxing threats, and parasocial

dynamics as recurring challenges. GREY INSIGHT ACM Digital Library

Parasocial relationships are the specific psychological risk in an NSFW photo-sharing community. Ballantine's research found that lurkers form one-sided emotional attachments to active posters. RSIS International In a small server where someone shares intimate photos and receives reactions from 30 members, the power dynamic is inverted from what scores suggest: the poster is actually performing emotional labor for the benefit of consumers. A scoring system that weights photo posting heavily could inadvertently create coercive pressure to share more intimate content than members are comfortable with.

The critical design rule: never weight NSFW content sharing higher than other forms of participation. A member who converses, reacts to others, and participates in discussions but never posts photos should score competitively with someone who posts photos but doesn't engage. This prevents the score from becoming a mechanism that pressures vulnerable sharing.

The ethics of scoring demand a specific architecture

Transparency framework: tell what, hide how

Research supports a hybrid approach:

- **Fully transparent:** What actions earn points (messages, reactions, replies, photos) and their relative value. Members should know that reciprocal engagement matters and that recent activity counts more than historical activity.
- **Semi-transparent:** "Recent activity counts more than older activity" without revealing exact half-life constants. "Engaging with many different members matters" without revealing the exact breadth threshold.
- **Fully opaque:** Anti-gaming mechanisms, exact normalization formulas, daily cap numbers, and the specific threshold below which someone becomes a removal candidate.

Stack Overflow's fully transparent system works because it operates at massive scale where individual gaming has minimal impact. Hacker News deliberately hides ranking factors because at smaller scale, transparency invites optimization. Your server is closer to Hacker News's situation.

Scoring should feel like a safety net, not a leaderboard

The overjustification effect (Deci et al., 1999) predicts that attaching external rewards to intrinsically motivated behavior *decreases* that motivation. Smiles at Work If members

currently share photos because they enjoy it, making photo-sharing a scored activity can transform enjoyment into obligation. Growth Engineering's research found that **public leaderboards disenfranchise the majority of users** — members who see themselves ranked low with no hope of catching up rationally disengage. [Growth Engineering](#)

Never create a visible leaderboard or public ranking. Never let members see each other's scores. Consider letting members check their own score privately if they ask, but don't push it. Frame the system as: "We track activity to identify members who have truly left, so we can make room for people on the waitlist." This is fundamentally different from "We rank members by quality."

Set the removal threshold generously. The goal is to identify the bottom 5–10% who are genuinely gone — haven't logged in for months, never react, never read. Not to create a competitive hierarchy among the active 90%.

The removal process matters more than the score

Research on how pruning affects remaining members is clear: watching others be removed creates **belonging uncertainty** — "Am I next?" Oxford Academic research found that the mere *perception* of surveillance creates self-censorship in 74% of people and reduced trust in 68%. [Academia.edu](#) In a micro-community where everyone knows each other, every removal is felt personally.

A graduated, humane process based on synthesized best practices:

1. **Automated flagging:** Bot identifies members below the activity threshold for 60+ consecutive days.
2. **Human review:** Admin reviews each flagged member's history. Was this person a pillar who recently went quiet? A new member still onboarding? Someone who mentioned personal difficulties?
3. **Personal outreach:** Admin sends a private, warm DM: "Hey, noticed you've been away — just checking in. No pressure, but wanted to make sure you know the door's open."
4. **14-day response window:** A single reaction emoji to the DM counts as "I'm here."
5. **Dormant role:** If no response, transition to a "dormant" role with reduced channel access but preserved membership. Explain this is reversible.
6. **Final removal:** After 30 additional days dormant with zero activity, remove with a kind message and an explicit open invitation to return anytime.
7. **Never remove more than 10–15% of the community at once.** Mass removals create panic.

8. **Never prune during community drama or known low-activity periods** (holidays, summer).
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Handling the edge cases that break naive algorithms

Leave of absence for life events

A purely algorithmic system cannot distinguish between disinterest and a death in the family. Implement a “leave of absence” feature: members DM an admin to flag they need a break, which pauses scoring for 30–90 days. This is lightweight to administer and prevents the worst outcomes. The 60-day minimum threshold before flagging already provides a buffer for most short-term life events.

Seasonal adjustment

Analyze your server’s activity patterns across several months before implementing scoring. If community-wide activity drops 40% in December, individual thresholds should drop proportionally. A simple approach: calculate the community’s average weekly activity as a baseline, and express individual scores as a percentage of the community average rather than an absolute number. This auto-adjusts for seasons.

The valuable quiet member

Distinguish between a **lurker** (logs in, reads messages, occasionally reacts) and a **ghost** (hasn’t opened Discord in months). Track presence signals — any evidence of reading, viewing, or reacting — as distinct from posting. A member who reacts to one message per week is clearly present and engaged, even if they never type a word. Set the bar at “are they actually here?” not “are they contributing content?” Research from Zhang and Storck (2001) found that peripheral members “benefit more from knowledge exchange and contribute as much as non-peripheral members.” [ScienceDirect](#)

New member grace period

Exempt new members from scoring for their first 30–60 days. Their clock starts after the grace period. During onboarding, communicate clearly what participation looks like in this community — not as a threat, but as a welcome: “Here’s how people typically engage here.”

Formerly active members who went quiet

A member who was a pillar for two years but went quiet for three months is fundamentally different from someone who joined, posted once, and disappeared. Implement a **legacy buffer**: long-term historical contribution provides a cushion against short-term inactivity.

One approach is to add a “tenure bonus” — members who have been active for 6+ months get an additional 30 days before being flagged. Members active 12+ months get 60 additional days. This respects the relationships and institutional knowledge that long-term members carry.

Conclusion: design for the community you want, not the metrics you can track

The deepest insight from this research is that **the scoring system will shape community culture more than it measures it**. Goodhart’s Law guarantees that whatever you measure, members will optimize for. If you weight photo posting, you’ll get more photos and less conversation. If you weight reactions, you’ll get emoji spam. If you weight replies, you’ll get “nice pic” comments. The only way to win this game is to measure so many dimensions simultaneously that gaming any single one provides minimal advantage — and to set thresholds low enough that genuine participants never think about the score at all.

The most successful community evaluation systems — Discourse trust levels, EPGP, private tracker ratios — share common DNA: they use **rolling time windows** (not cumulative scores), **require breadth** (unique users, unique days, unique channels), **include decay** (recent activity matters more), and **separate quality from quantity**. They work because they align incentives: the easiest way to score well is to actually participate, and the system is forgiving enough that normal human variability doesn’t trigger punishment.

For an NSFW community specifically, the ethical imperative is clear: the score must never create pressure to share intimate content, individual scores must remain private, removal must follow a humane graduated process, and the system should be explained to the community before implementation — with their input shaping the final design. The goal is not to rank members from best to worst. It is to identify the small number of genuinely absent members and make room for people who want to be there. Frame it that way, build it that way, and the community will be stronger for it.